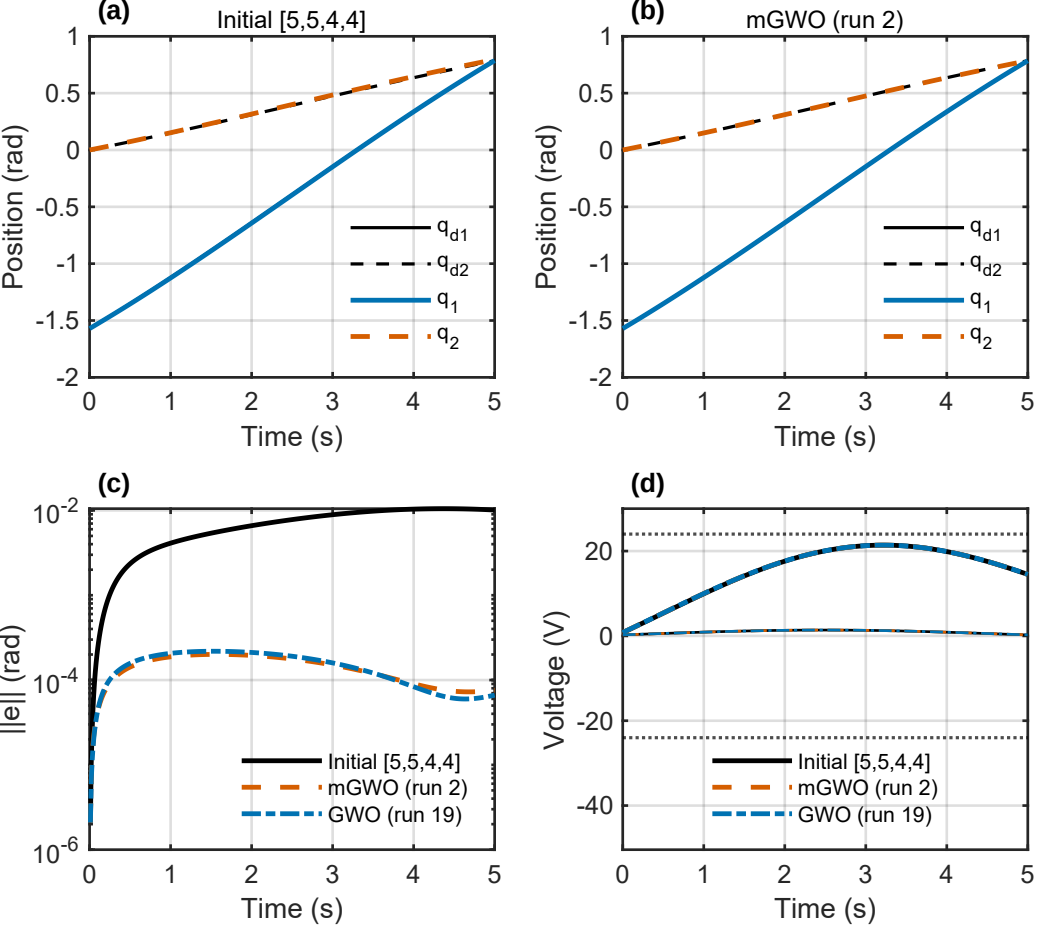

(a)
Initial [5,5,4,4]
Position (rad)
Time (s)
q_{d1}
q_{d2}
q_1
q_2
(b)
mGWO (run 2)
Position (rad)
Time (s)
q_{d1}
q_{d2}
q_1
q_2
(c)
||e|| (rad)
Time (s)
Initial [5,5,4,4]
mGWO (run 2)
GWO (run 19)
(d)
Voltage (V)
Time (s)
Initial [5,5,4,4]
mGWO (run 2)
GWO (run 19)